

22.1 A 1.1V 16GB 640GB/s HBM2E DRAM with a Data-Bus Window-Extension Technique and a Synergetic On-Die ECC Scheme

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Rapidly evolving artificial intelligence (AI) technology, such as deep learning, has been successfully deployed in various applications: such as image recognition, health care, and autonomous driving. Such rapid evolution and successful deployment of AI technology have been possible owing to the emergence of accelerators, such as GPUs and TPUs, that have a higher data throughput. This, in turn, requires an enhanced memory system with large capacity and high bandwidth [1]; HBM has been the most preferred high-bandwidth memory technology due to its high-speed and low-power characteristics, and 1024 I/Os facilitated by 2.5D silicon interposer technology, as well as large capacity realized by through-silicon via (TSV) stack technology [2]. Previous-generation HBM2 supports 8GB capacity with a stack of 8 DRAM dies (i.e., 8-high stack) and 341GB/s (2.7Gb/s/pin) bandwidth [3]. The HBM industry trend has been a speed improvement of 15~20% every year, while capacity increases by 1.5~2 $\times$ every two years. In this paper, we present a 16GB HBM2E with circuit and design techniques to increase its bandwidth up to 640GB/s (5Gb/s/pin), while providing stable bit-cell operation in the 2nd generation of a 10nm DRAM process: featuring (1) a data-bus window-extension technique to cope with reduced t_{CCD} , (2) a power delivery network (PDN) designed for stable operation at a high speed, (3) a synergetic on-die ECC scheme to reliably provide large capacity, and (4) an MBIST solution to efficiently test large capacity memory at a high speed.

Figure 22.1.1 shows the chip and bus architectures of HBM2 [4] and HBM2E presented in this work. The larger memory capacity and higher operating frequency at fixed t_{CCD} (2nCK) increase signal skews while decreasing the timing window of data-line fetch. The location of TSVs to minimize the length of data lines on the buffer die also makes the signal skews worse for the core dies. In HBM2 [4], the bank group bus (BG_BUS) with 2nCK fetch timing window was adopted to implement both BL2 legacy and BL4 pseudo-channel mode operations. For bank group operation, the timing window of global bus (GBUS) and vertical bus (VBUS) is 1nCK. The maximum bandwidth of HBM2 was limited to 2.4Gb/s/pin due to the GBUS signal skews and the read/write cell operation cycle time of 2nCK. In recent work, the bus architecture of both BG_BUS and GBUS with 2nCK is proposed for 2.7Gb/s/pin operation [3].

In this HBM2E design, the timing window of BG_BUS is extended from 2nCK to 4nCK with BL4 (burst length of 4) pseudo-channel mode only operation. Accordingly, the read/write cell operation cycle time is widened to 4nCK (t_{CCD}) and 256 BG_BUS with 4nCK timing window in each bank group are placed on array without increasing array area although the number of bus lines is doubled. GBUS with a 2nCK timing window is placed across the TSV area to eliminate the redundant GBUS routing area in the previous HBM [4]. The block and timing diagrams of the data bus control is described in Fig. 22.1.2. The proposed data bus window of BG_BUS and GBUS is increased by 2 $\times$ to achieve a high bandwidth operation up to 5.0Gb/s/pin.

In contrast to commodity DRAM, HBMs consume a large amount of power since multiple channels operate simultaneously in a 3D-stacked structure. When there is a considerable IR drop, skews occur among data bus signals, resulting in performance degradation. In order to overcome this issue, top metal layer is dedicated to power routing in this chip design. In order to maximize the benefit of the power-exclusive top metal usage, power TSVs are placed at the chip edge and in the middle of cell array regions which are vulnerable to IR drop in core die. Edge power TSVs are connected to the adjacent back-bone power line using a lower metal layer as shown in Fig. 22.1.3. Edge power TSVs are vulnerable to

power short since the amount of deviation is the largest when there is a z-axis direction misalignment during chip on wafer (CoW) stacking process. In order to cope with this situation, align monitoring bumps are placed at four edges of a core die. With these proposed power integrity design, the IR drop of core dies is improved by 62%, enabling stable high frequency HBM2E operation up to 640GB/s.

HBM provides additional memory cells for system ECC parity bits in order to achieve high quality at a high temperature (e.g., 95°C). During read or write operation, 256b data as well as associated 32 parity bits for system ECC are accessed. With this 256b data and 32 parity bits, a (72, 64) SECDED ECC scheme can be applied, as shown in Fig. 22.1.4. To further enhance the reliability, an on-die ECC (OD-ECC) scheme is implemented. More specifically, after a block of 256b data is divided into three sub-blocks of 96b data, a (104, 96) SECDED scheme is applied to each sub-block, which makes the on-die ECC scheme orthogonal to the system ECC scheme. This can reduce uncorrectable bit error rate (UBER) by 3 times; random double-bit errors in any code-words of system ECC can be corrected by OD-ECC, if the double-bit errors are located in two OD-ECC code-words (e.g., one-bit error in OD-ECC 0 and the other in OD-ECC 1). The comparison of BERs among three schemes: (1) no ECC, (2) only system ECC, and (3) system ECC with OD-ECC is shown in Fig. 22.1.4 (bottom). During write operation, written data from I/O pins are buffered and parities are encoded. After that, they are resynchronized and written into cells. During read operation, data read from cells are buffered and syndromes are calculated. If there is 1b error in one OD-ECC code-word, this is corrected. After correction, corrected data and all other data are re-synchronized and delivered to I/O pins.

HBM2E also has a memory built-in self-test (MBIST) function to screen manufacturing defects occurred during system-in package (SiP) fabrication as illustrated in Fig. 22.1.5. The low frequency (LF) mode screens defects of cell and core with specified test patterns and two functions are implemented to detect defects: (1) a conventional DDR test and (2) the proposed parallel-bit test (PBT). The DDR test compares two sets of data, one is read data from a core die and the other is expected data generated by the MBIST engine. As the memory capacity increases, the required time for memory cell test increases proportionally. HBM2E has a 16Gb/2-channel capacity, which is doubled from the previous HBM2, increasing test time by 2 $\times$. In order to reduce the test time, a PBT mode is proposed in this design. The PBT mode accesses 8 banks with one command. The test results from the 8 banks are compared at the core die. If a failure is detected after the comparison, the fail address is stored in fail address memory (FAM) in the buffer die. After finishing MBIST operation, FAM data can be accessed via the IEEE1500 interface. The proposed PBT mode reduces test time by 65 % compared to the DDR mode.

The high frequency (HF) mode generates an IDD7 pattern and having a method to adjust AC parameters such as t_{RCD} , t_{CCD} and t_{RRD} according to the operating frequency. For design simplicity, MBIST HF engine generates 8 parallel commands and data at low operating frequency such as 312.5MHz. After that, Serializer converts parallel low frequency commands and data of 312.5MHz to high frequency ones of 2.5GHz. High frequency CLK source is required for serializer and this can be supplied via on-chip PLL or external CLK from PHY ball. During read operation, one channel high frequency data are multiplexed and their frequency is lowered by 0.125 $\times$ at de-serializer block. The de-serialized data are compared with expected data and results are accessed by the IEEE1500 interface.

Figure 22.1.6(left) shows the measured t_{CK} shmoo. The fabricated HBM2E achieves 5Gb/s/pin at 1.1V. The key features are summarized in Fig. 22.1.6(right). The chip micrograph of the fabricated core and buffer dies are shown in Fig. 22.1.7.

References:

- [1] N. P. Jouppi et al., "In-Datcenter Performance Analysis of a Tensor Processing Unit," ISCA, 2017.
- [2] D. U. Lee et al., "A 1.2V 8Gb 8-Channel 128GB/s High-Bandwidth Memory (HBM) Stacked DRAM with Effective Microbump I/O Test Methods Using 29nm Process and TSV," *ISSCC*, pp. 432-433, Feb. 2014.
- [3] J. H. Cho et al., "A 1.2V 64Gb 341GB/s HBM2 Stacked DRAM with Spiral Point-to-Point TSV Structure and Improved Bank Group Data Control," *ISSCC*, pp. 208-209, 2018.
- [4] K. Sohn et al., "A 1.2V 20nm 307GB/s HBM DRAM with At-Speed Wafer-Level I/O Test Scheme and Adaptive Refresh Considering Temperature Distribution," *ISSCC*, pp. 316-317, Feb. 2016.

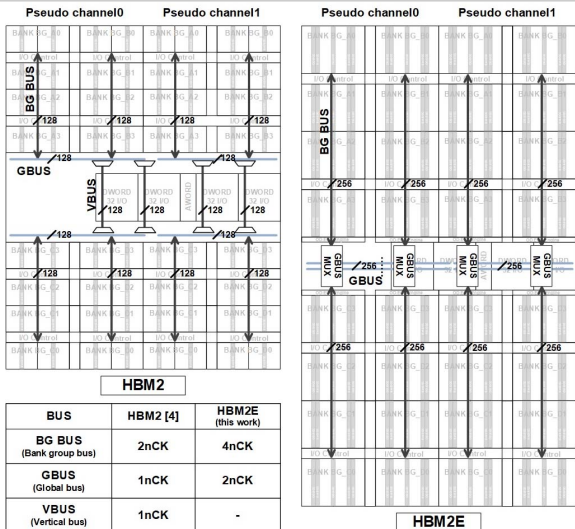


Figure 22.1.1: Chip and bus architecture comparison between 4Gb/1-channel HBM2 [4] and 8Gb/1-channel HBM2E.

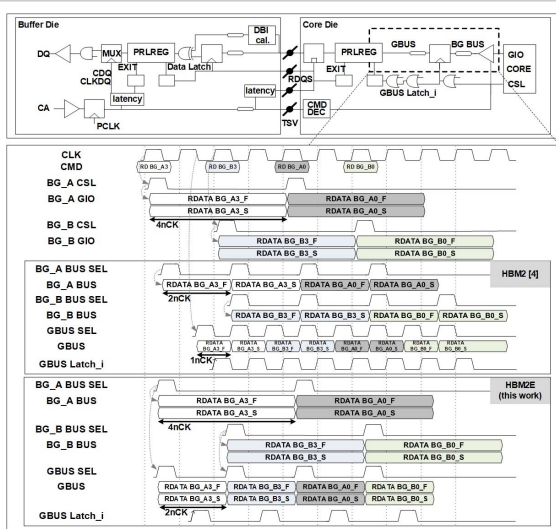


Figure 22.1.2: Block and timing diagram of the data bus control for HBM2 [4] and HBM2E.

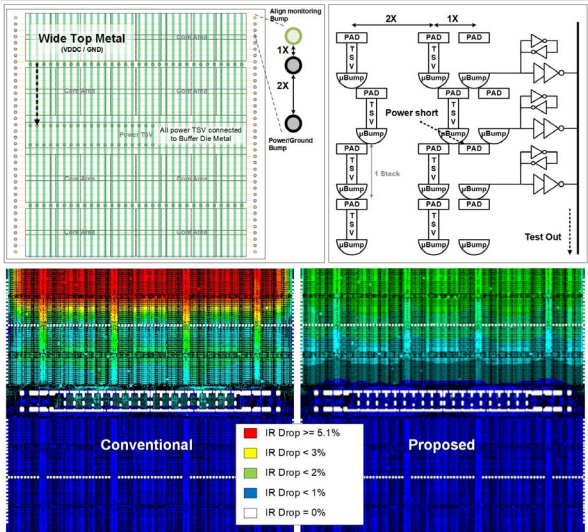


Figure 22.1.3: On-chip power integrity design utilizing a wide top metal dedicated to power mesh for a stable 640GB/s high speed operation.

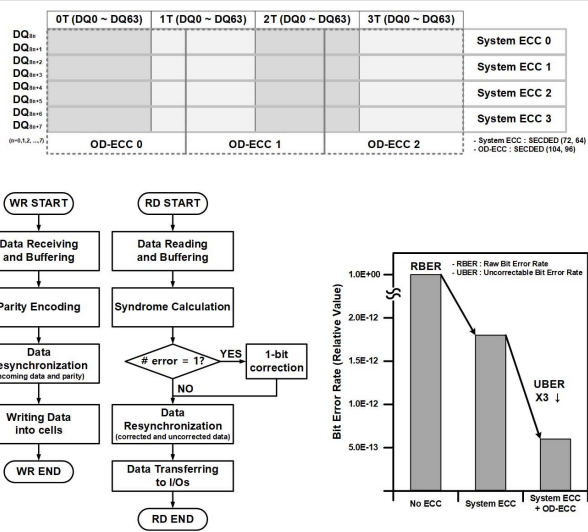


Figure 22.1.4: Synergetic On-Die ECC scheme with an orthogonal ECC code to system-ECC.

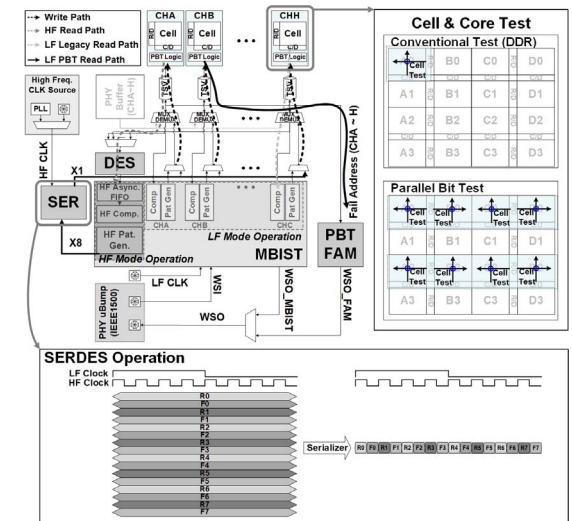


Figure 22.1.5: Block diagram of the proposed memory built-in self-test (MBIST) featuring a fast PBT and HF mode.

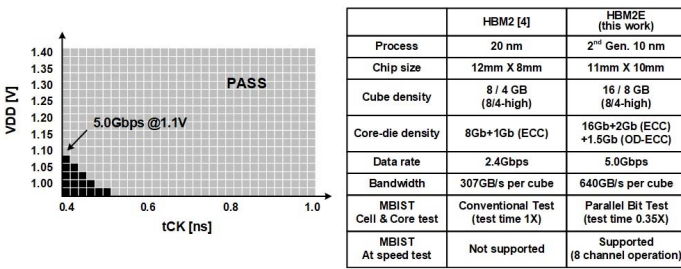


Figure 22.1.6: Measured t_{ck} shmoo (left) and key feature summary (right).

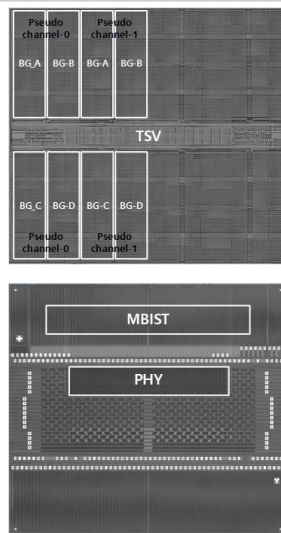


Figure 22.1.7: Chip micrograph of 16Gb/2-channel core die (top) and buffer die (bottom) fabricated in the 2nd generation of a DRAM process.